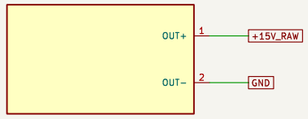
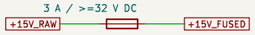
anden-dcc / ONE OUTPUT, TWO TRACKS

Pico W + Deek-Robot L298P + USB-C PD + MP1584.

M3
USB-C PD / 15 V, >=3 A

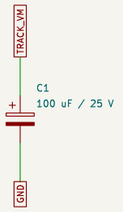


F1
3 A / >=32 V DC



S1

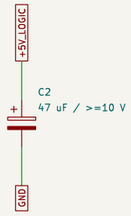




S1: manual motor-power cutoff.
Logic stays powered when S1 is off.
Use contacts rated >=30 V DC / 3 A.

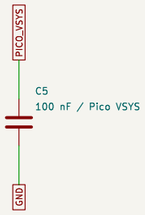
M4
MP1584 / SET 5.00 V

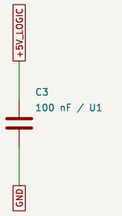


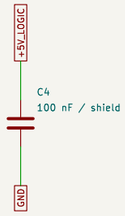


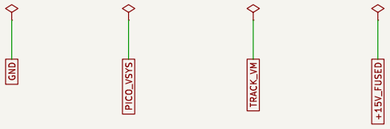
D1
1N5819 / band to VSYS





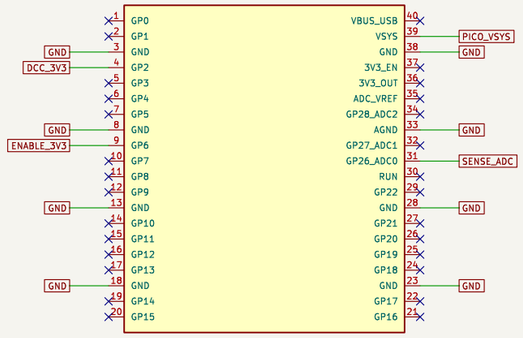




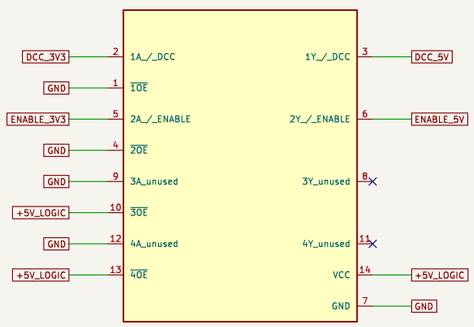


USB recovery: external 5 V goes through D1 to VSYS pin 39.
Leave VBUS and 3V3_OUT isolated from the external 5 V supply.

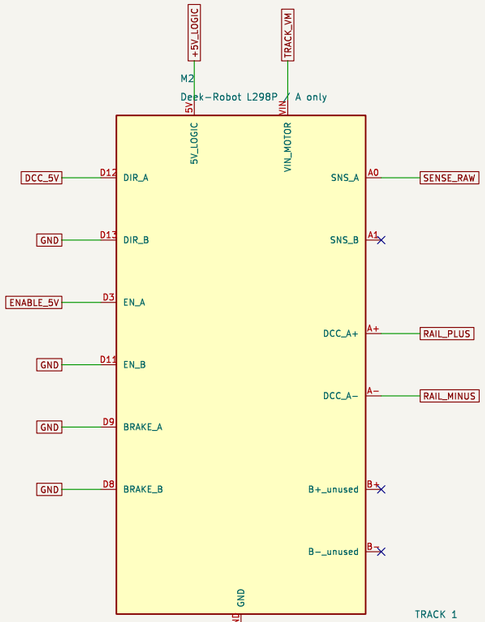
M1
Raspberry Pi Pico W



U1
SN74AHCT125N



M2
Deek-Robot L298P / A only



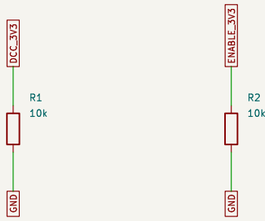
J1
TRACK 1



J2
TRACK 2



GP26: nominal 0.825 V/A after divider.
Calibrate before enabling track power.









A+ feeds one rail of BOTH tracks.
A- feeds their other rails. Neither rail is GND.
Each loco is still controlled by its own DCC address.
Channel B is unused; do not parallel A and B outputs.
Power enable / current cutoff are not in firmware yet.

Firmware 0.5.2: GP2 logic only; track power disabled
NOT BUILT OR ELECTRICALLY VALIDATED
anden-dcc / laullon

Sheet: /
File: anden-dcc.kicad_sch

Title: anden-dcc / simple one-output wiring

Size: A3	Date: 2026-10-29	**Rev: C**
KiCad E.D.A. 8.0.6		Id: 1/1